

## Frozen representations or models

## Trainable interface or head

### global-embedding heads

frozen image global vector (512)  
frozen question global vector (512)

linear/ReLU/dropout/linear head  
over the selected global-vector  
composition

### token-level latent reasoner

50 frozen image tokens: CLS plus  
49 patches, width 512  
variable-length frozen question  
tokens, width 512

32 learned latent queries; 4  
blocks at width 512 with 8  
attention heads; question cross-  
attention, image cross-attention,  
self-attention and feed-forward  
processing

### question-side SLM interface

CLIP question-token store for A0p  
pretrained or matched-random  
SmolLM2-135M for A1/A1r

A0p Linear(512, 512)  
A1/A1r Linear(576, 512)  
latent reasoner and classifier

Projection to common width 512 is an interface operation and is not  
evidence of a naturally shared pretrained CLIP/SmolLM2 semantic space.

### answer-side SLM interface

matched-random or pretrained  
SmolLM2-135M for B2/B3  
matched-random or pretrained  
SmolLM2-360M for B4r/B4

reasoner trunk  
projection to language-model width  
B1 attention-pool classifier where  
applicable

B1 language-model-free classifier  
R1 candidate-answer likelihood ranking  
R2 trie-constrained greedy  
R3 bounded free greedy

### compact-VLM contextual class

SmolVLM-256M-Instruct  
SmolVLM-500M-Instruct  
entire integrated VLM

Trainable components: none

evaluation-only contextual systems; not recipe-matched trained arms